FORM 2

THE PATENTS ACT, 1970

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See sections 10 & rule 13)

1. TITLE OF THE INVENTION

SYSTEM AND METHOD FOR 3D ULTRASOUND IMAGING OF TARGET ANATOMY

2. APPLICANT (S)

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3. PREAMBLE TO THE DESCRIPTION

COMPLETE SPECIFICATION

The following specification particularly describes the invention and the manner in which it is to be performed.

TECHNICAL FIELD

[0001] The present disclosure generally relates to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional (3-D) ultrasound volumetric reconstruction.

BACKGROUND

[0002] Ultrasound imaging is widely used across clinical applications due to its non-ionizing nature, real-time acquisition capability, and portability. Conventional two-dimensional (2-D) ultrasound is commonly employed for anatomical visualization and diagnostic decision-making; however, its planar perspective may limit assessment of spatial relationships within complex anatomical structures. Three-dimensional (3-D) ultrasound volumetric reconstruction therefore offers significant advantages by enabling depth-resolved visualization, quantitative analysis, and improved anatomical interpretation.

[0003] Current approaches for 3-D ultrasound reconstruction often require additional hardware to determine the spatial position of the ultrasound probe during acquisition. External tracking systems, such as electromagnetic, optical, or mechanical encoders, are commonly employed to localize the probe. These solutions may increase system cost, add mechanical bulk, impose line-of-sight constraints, or introduce sensitivity to surrounding electromagnetic environments. In addition, the need for calibration and maintenance of such tracking systems may limit their practicality in fast-paced or resource-constrained clinical settings.

[0004] Other techniques attempt to estimate probe position from acquired ultrasound signals alone; however, such methods may be computationally intensive or insufficiently robust, especially when imaging anatomies with limited textural features. Consequently, there remains a challenge in performing reliable 3-D ultrasound volumetric reconstruction using a conventional 2-D ultrasound probe without employing dedicated external tracking devices. Furthermore, existing tracking attachments for 3-D reconstruction may rely on moving mechanical components, which may introduce wear, alignment drift, or operational failure over repeated use. Such mechanical limitations may reduce long-term consistency, increase maintenance overhead, and complicate integration into standard clinical workflows.

[0005] Therefore, there is a need for an improved approach that overcomes at least the above-mentioned drawbacks.

OBJECTS OF THE PRESENT DISCLOSURE

[0006] A general object of the present disclosure is to provide a system and a method for three-dimensional (3-D) ultrasound volumetric reconstruction of a target anatomy using a conventional two-dimensional (2-D) ultrasound probe.

[0007] Another object of the present disclosure is to enable sensorless localization of the ultrasound probe during volumetric acquisition without requiring external tracking systems.

[0008] Another object of the present disclosure is to utilize a reflector-integrated mask to produce a reflector trace within captured ultrasound frames, thereby facilitating determination of spatial frame positions.

[0009] Another object of the present disclosure is to achieve low mechanical and computational complexity while maintaining compatibility with existing ultrasound imaging workflows.

[0010] Another object of the present disclosure is to generate accurate three-dimensional ultrasound reconstructions in a cost-effective manner suitable for use in diverse clinical environments, including resource-constrained settings.

SUMMARY

[0011] Aspects of the present disclosure pertain to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional (3-D) ultrasound volumetric reconstruction.

[0012] An aspect of the present disclosure pertains to a reflector-integrated mask (RIM) for ultrasound imaging. The RIM includes an outer scan track defined by a rigid, four-walled, open-ended rectangular cuboid housing. The RIM also includes an inner scan track disposed within the outer scan track, wherein the inner scan track is dimensioned to receive a linear ultrasound probe and to constrain translation of the linear ultrasound probe along a scan direction. The RIM also includes a linear acoustic reflector fixed to or embedded in a side wall of the inner scan track and extending along the scan direction. Herein, the linear acoustic reflector is disposed in a slanted, non-parallel orientation with respect to an imaging plane of the received linear ultrasound probe and positioned to lie within a field of view of the imaging plane during the translation.

[0013] In one embodiment, wherein the linear acoustic reflector includes any metallic object e.g., stainless steel, plastic object or polylactic acid (PLA) based object, but not limited thereto.

[0014] In one embodiment, the RIM further includes an acoustic-coupling medium disposed within the inner scan track.

[0015] In one embodiment, a protrusion width (WR) of the linear acoustic reflector is an integer multiple of an element size of the linear ultrasound probe.

[0016] An aspect of the present disclosure pertains to a system for ultrasound imaging. The system includes a linear ultrasound probe configured to acquire a plurality of two-dimensional (2-D) ultrasound frames of a target anatomy. The system also includes a reflector-integrated mask (RIM) that includes an inner scan track dimensioned to receive the linear ultrasound probe and to constrain translation of the linear ultrasound probe along a scan direction of the target anatomy. The RIM also includes a linear acoustic reflector fixed to or embedded in a side wall of the inner scan track and extending along the scan direction. Herein, the linear acoustic reflector is disposed in a slanted, non-parallel orientation with respect to an imaging plane of the linear ultrasound probe and positioned to lie within a field of view of the imaging plane during said translation. The system also includes a memory and a processor. The processor is operatively coupled to the linear ultrasound probe and the memory, wherein the memory stores instructions that when executed by the processor, cause the processor to receive, from the linear ultrasound probe, the plurality of 2-D ultrasound frames acquired along the scan direction. The processor is also configured to identify, in each 2-D ultrasound frame, a reflector trace corresponding to the linear acoustic reflector. The processor is also configured to determine, based on the reflector trace and a calibration parameter stored in the memory, a relative lateral position of each 2-D ultrasound frame along the scan direction. The processor is also configured to generate, using the plurality of 2-D ultrasound frames and the respective lateral positions, the 3-D volumetric reconstruction of the target anatomy.

[0017] In one embodiment, the stored calibration parameter includes a reflector angle and a reflector offset determined during initial two-point static calibration.

[0018] In one embodiment, the processor is configured to compute the relative lateral position of each 2-D ultrasound frame according to a closed-form geometric relationship based on the pixel-depth of the reflector trace.

[0019] In one embodiment, the processor is configured to order the plurality of 2-D ultrasound frames based on their respective relative lateral positions along the scan direction prior to generation of the 3-D volumetric reconstruction.

[0020] In one embodiment, the processor is configured to generate the 3-D volumetric reconstruction using delay-and-sum (DAS) beamforming applied to the plurality of 2-D ultrasound frames.

[0021] Another aspect of the present disclosure pertains to a method for three-dimensional (3-D) ultrasound volumetric reconstruction. The method includes acquiring, by the linear ultrasound probe, a plurality of two-dimensional (2-D) ultrasound frames of a target anatomy by constrained translation of the linear ultrasound probe, within an inner scan track of a reflector-integrated mask (RIM), along a scan direction of the target anatomy. The method also includes receiving, by a processor operatively coupled to a memory, the plurality of 2-D ultrasound frames acquired along the scan direction. The method also includes identifying, by the processor, in each 2-D ultrasound frame, a reflector trace corresponding to the linear acoustic reflector of the RIM positioned to lie within a field of view of an imaging plane of the linear ultrasound probe during the translation, wherein the linear acoustic reflector is disposed in a slanted, non-parallel orientation with respect to the imaging plane. The method also includes accessing, by the processor, a calibration parameter stored in the memory. The method also includes determining, by the processor, based on the reflector trace and the calibration parameter, a relative lateral position of each 2-D ultrasound frame along the scan direction. The method also includes generating, by the processor, using the plurality of 2-D ultrasound frames and their respective relative lateral positions, a 3-D volumetric reconstruction of the target anatomy.

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The accompanying drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain the principles of the present disclosure.

[0023] FIG. 1 illustrates an exemplary overview of a system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction, in accordance with an embodiment of the present disclosure.

[0024] FIG. 2 illustrates an exemplary isometric diagram of the reflector-integrated mask (RIM) (104) positioned above a target anatomy (204), in accordance with an embodiment of the present disclosure.

[0025] FIG. 3 illustrates an exemplary exploded view of assembly of the reflector-integrated mask (RIM) (104), in accordance with an embodiment of the present disclosure.

[0026] FIG. 4 illustrates an exemplary block diagram of the system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction, in accordance with an embodiment of the present disclosure.

[0027] FIG. 5 illustrates an exemplary schematic diagram of the RIM (104) in Y-axis direction, in accordance with an embodiment of the present disclosure.

[0028] FIG. 6 illustrates an exemplary block diagram of the processing unit (120), in accordance with an embodiment of the present disclosure.

[0029] FIG. 7 illustrates an exemplary flow diagram of a method (700) for three-dimensional (3-D) ultrasound volumetric reconstruction, in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0030] In the following description, for the purposes of explanation, various specific details are set forth in order to provide a thorough understanding of embodiments of the present disclosure. It will be apparent, however, that embodiments of the present disclosure may be practiced without these specific details. Several features described hereafter can each be used independently of one another or with any combination of other features. An individual feature may not address all of the problems discussed above or might address only some of the problems discussed above. Some of the problems discussed above might not be fully addressed by any of the features described herein.

[0031] The ensuing description provides exemplary embodiments only and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the exemplary embodiments will provide those skilled in the art with an enabling description for implementing an exemplary embodiment. It should be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the disclosure as set forth.

[0032] The following is a detailed description of embodiments of the disclosure depicted in the accompanying drawings. The embodiments are in such detail as to clearly communicate the disclosure. If the specification states a component or feature “may”, “can”, “could”, or “might” be included or have a characteristic, that particular component or feature is not required to be included or have the characteristic.

[0033] As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

[0034] Conventional three-dimensional ultrasound reconstruction typically requires knowledge of the spatial position of a two-dimensional ultrasound probe during data acquisition. Existing systems often rely on external optical, electromagnetic, or mechanical tracking devices to localize the probe, which increases system complexity, cost, and susceptibility to misalignment or environmental interference. These additional tracking components may introduce bulky hardware, require repeated calibration, and limit portability, making them less suitable for resource-constrained environments and bedside imaging. Moreover, tracking-free reconstruction approaches based solely on image content may be unreliable when anatomical structures lack sufficient texture, resulting in inaccurate frame ordering and suboptimal volumetric reconstruction.

[0035] To address these challenges, the present disclosure provides a system and a method for three-dimensional ultrasound volumetric reconstruction that employ a reflector-integrated mask to facilitate sensorless localization of a two-dimensional ultrasound probe. The reflector-integrated mask incorporates a linear acoustic reflector that is positioned at a slanted, non-parallel orientation relative to the probe's imaging plane so that a reflector trace appears consistently within each acquired frame. A processing unit identifies the reflector trace, utilizes a stored calibration parameter, determines the relative lateral position of each acquired frame, and generates a three-dimensional volumetric reconstruction. This approach enables position encoding directly within the acquired ultrasound data, eliminating the need for external tracking hardware.

[0036] By embedding positional information into the captured ultrasound frames through the reflector-generated trace, the disclosed system enables accurate spatial ordering of image data without reliance on external tracking devices. This reduces mechanical and computational overhead, leading to improved portability, lower system cost, and simplified maintenance. The ability to generate volumetric reconstructions using a standard two-dimensional ultrasound probe enhances accessibility in diverse clinical settings, including point-of-care and resource-limited environments. The resulting volumetric representation may provide improved anatomical visualization, aiding diagnostic decision-making while maintaining compatibility with conventional ultrasound acquisition workflows.

[0037] The present disclosure pertains to the field of medical imaging technology. In particular, the present disclosure relates to a system and a method for three-dimensional (3-D) ultrasound volumetric reconstruction.

[0038] Various embodiments with respect to the present disclosure will be explained in detail with reference to FIGs. 1-7.

[0039] Referring now to FIG. 1, an exemplary overview of a system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction is illustrated, in accordance with one or more embodiments of the present disclosure. The system (100) may include a linear ultrasound probe (102), a reflector-integrated mask (RIM) (104), a processing unit (120), a plurality of two-dimensional (2-D) ultrasound frames (114), and a 3-D volumetric reconstruction (116).

[0040] In one embodiment, the linear ultrasound probe (102) may be positioned within an outer scan track (106) of the reflector-integrated mask (RIM) (104) and guided along an inner scan track (108). The inner scan track (108) may be dimensioned to receive the linear ultrasound probe (102) and to constrain translation of the linear ultrasound probe (102) along a scan direction (Y) during imaging of a target anatomy.

[0041] The reflector-integrated mask (RIM) (104) may include a linear acoustic reflector (110) disposed along the inner scan track (108). The linear acoustic reflector (110) may be positioned within a field-of-view of an imaging plane (112) of the linear ultrasound probe (102), such that a reflector trace (118) may be observable in each of the plurality of 2-D ultrasound frames (114) (also referred to as 2-D B-mode image) as the linear ultrasound probe (102) translates along the scan direction (Y).

[0042] As the linear ultrasound probe (102) is translated along the scan direction (Y), the imaging plane (112) may acquire the plurality of 2-D ultrasound frames (114) of the target anatomy. The reflector trace (118) captured within each of the plurality of 2-D ultrasound frames (114) may enable subsequent determination of relative frame positions for volumetric reconstruction.

[0043] The plurality of 2-D ultrasound frames (114) may be provided to the processing unit (120), which may be configured to process the plurality of 2-D ultrasound frames (114) and generate the 3-D volumetric reconstruction (116). The sequence at the right portion of FIG. 1 illustrates an example of the plurality of 2-D ultrasound frames (114) being spatially organized based on their relative positions to form the 3-D volumetric reconstruction (116) (referred to as 3D B-mode image) of the target anatomy.

[0044] Referring to FIG. 2, an exemplary isometric diagram of the reflector-integrated mask (RIM) (104) positioned above a target anatomy (204) is illustrated, in accordance with one or more embodiments of the present disclosure.

[0045] An ultrasound sensor array (202) of the linear ultrasound probe (102) may be aligned above the inner scan track (108) such that the ultrasound sensor array (202) may face downward toward the linear acoustic reflector (110) and the target anatomy (204). In operation, the ultrasound sensor array (202) may be positioned to acquire ultrasound echoes along a field-of-view bounded by the imaging plane (112) when the linear ultrasound probe (102) translates along the scan direction (Y).

[0046] The reflector-integrated mask (RIM) (104) may be placed in acoustic contact with subject skin (206) overlying the target anatomy (204). During scanning, the ultrasound sensor array (202) may emit and receive ultrasound signals that propagate through the subject skin (206) to image the target anatomy (204). The linear acoustic reflector (110) disposed within the inner scan track (108) may generate a reflector trace (118) within the acquired 2-D ultrasound frames (114), corresponding to the relative position of the linear ultrasound probe (102) along the scan direction (Y).

[0047] The coordinate axes (X, Y, Z) shown in FIG. 2 indicate the imaging plane (112) as XZ, and the scan direction as Y associated with translation of the linear ultrasound probe (102) and imaging performed within the reflector-integrated mask (RIM) (104).

[0048] Referring to FIG. 3, an exemplary exploded view of assembly of the reflector-integrated mask (RIM) (104), in accordance with one or more embodiments of the present disclosure.

[0049] In one embodiment, the reflector-integrated mask (RIM) (104) may include an outer scan track (106) defined by a rigid, four-walled, open-ended rectangular cuboid housing, and an inner scan track (108) disposed within the outer scan track (106). The inner scan track (108) may be dimensioned to receive a linear ultrasound probe and to constrain translation of the probe along a scan direction. A linear acoustic reflector (110) may be embedded in or fixed to a side wall of the inner scan track (108) and may extend along the scan direction.

[0050] In some embodiments, the linear acoustic reflector (110) may stretch out within close proximity of the walls of the inner scan track (108), and may be disposed in a slanted, non-parallel orientation with respect to the imaging plane of the received linear ultrasound probe, such that the linear acoustic reflector (110) may remain positioned within a field-of-view during translation.

[0051] In an alternative configuration, the inner scan track (108) may be in a form of a continuous protrusion extending toward the walls of the outer scan track (106), with a predetermined protrusion height selected to support constrained translation of the linear ultrasound probe along the scan direction of the target anatomy. The ultrasound sensor array (202) may be positioned above the inner scan track (108) such that, after assembly, the ultrasound sensor array (202) may be aligned to view both the linear acoustic reflector (110) and underlying tissue during scanning.

[0052] The linear acoustic reflector (110) may be positioned in a slanted, non-parallel orientation with respect to the imaging plane of the received linear ultrasound probe to ensure that the linear acoustic reflector (110) remains visible within the field-of-view throughout translation along the scan direction. A parallel arrangement would produce a reflector trace that remains at a constant depth, making it difficult to distinguish frame-to-frame probe motion from tissue depth variations. In contrast, the slanted configuration allows the reflector trace to appear at a progressively changing pixel-depth in sequential two-dimensional (2-D) ultrasound frames, thereby encoding positional information along the scan direction. This depth-dependent variation may be reliably extracted from each ultrasound frame to determine the relative lateral position of the probe, enabling accurate spatial ordering of frames for three-dimensional (3-D) reconstruction. The inclination further improves robustness against out-of-plane motion and ensures that at least a portion of the linear acoustic reflector (110) remains within the field-of-view even under slight translation or tilt of the linear ultrasound probe.

[0053] In one embodiment, the linear acoustic reflector (110) may include stainless steel, but not limited thereto. Stainless steel may provide a high acoustic-impedance contrast relative to surrounding polymers of the reflector-integrated mask (RIM) (104), resulting in a strong, repeatable specular return that forms a stable reflector trace (118) within the two-dimensional (2-D) ultrasound frames (114). Stainless steel may further offer dimensional rigidity, corrosion resistance in the presence of acoustic-coupling media, and compatibility with repeated cleaning or sterilization, thereby maintaining the reflector geometry and surface quality over extended use. In certain implementations, the linear acoustic reflector (110) may be fabricated as a strip or bar of stainless steel and fixed to or embedded in a side wall of the inner scan track (108) so as to extend along the scan direction (Y). A polished or uniformly finished stainless-steel surface may be selected to provide a consistent echo response across the length of the linear acoustic reflector (110), supporting reliable depth extraction and subsequent position determination during three-dimensional (3-D) volumetric reconstruction.

[0054] In one embodiment, the reflector-integrated mask (RIM) (104) may further include an acoustic-coupling medium disposed within the inner scan track (108). The acoustic-coupling medium may facilitate efficient transmission of ultrasound energy between the linear ultrasound probe (102) and the target anatomy (204) by reducing acoustic impedance mismatch and minimizing reflection losses at intervening boundaries. The acoustic-coupling medium may also promote consistent contact across the inner scan track (108) during translation of the linear ultrasound probe (102), thereby helping maintain signal quality along the scan direction (Y). In certain implementations, the acoustic-coupling medium may fill at least a portion of the inner scan track (108) so that the ultrasound sensor array (202) remains acoustically coupled without requiring continuous manual gel reapplication, enabling repeatable imaging conditions for generation of the plurality of two-dimensional (2-D) ultrasound frames (114) used in three-dimensional (3-D) volumetric reconstruction.

[0055] In one exemplary embodiment, examples of the acoustic-coupling medium disposed within the inner scan track (108) may include one or more ultrasound-compatible materials such as aqueous gel, hydrogel, saline solution, water, or other biocompatible fluids selected to provide stable acoustic impedance matching. In certain configurations, the acoustic-coupling medium may additionally include semi-solid or gel-based formulations designed to retain shape within the reflector-integrated mask (RIM) (104) to support reliable probe translation along the scan direction (Y).

[0056] In one embodiment, a protrusion width (WR) (as shown in FIG. 3) of the linear acoustic reflector (110) may be selected as an integer multiple of an element size of the linear ultrasound probe (102) (e.g., a pitch or active aperture dimension of an element of an ultrasound sensor array (202)). The linear acoustic reflector (110) may extend into the inner scan track (108) with the protrusion width (WR) aligned to the discrete sampling geometry of the linear ultrasound probe (102) so that the reflector trace (118) remains consistently detectable across lateral positions along the scan direction (Y).

[0057] Selecting the protrusion width (WR) as an integer multiple of the element size may help maintain a stable echo amplitude and a uniform apparent line thickness in the two-dimensional (2-D) ultrasound frames (114), reduce flicker or partial-element coupling effects during translation, and improve repeatability of depth extraction used for position determination. This sizing may also simplify manufacturing tolerance windows and alignment during assembly, thereby supporting reliable formation of the 3-D volumetric reconstruction (116) from frames acquired along the scan direction (Y).

[0058] Referring now to FIG. 4, an exemplary block diagram of the system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction is illustrated, in accordance with one or more embodiments of the present disclosure. The system (100) may include the linear ultrasound probe (102) configured to acquire the plurality of two-dimensional (2-D) ultrasound frames (114) of the target anatomy (204) during translation along the scan direction (Y) supported by the reflector-integrated mask (RIM) (104), as described previously with reference to FIGs. 1–3.

[0059] As further shown in FIG. 4, the linear ultrasound probe (102) may be operatively coupled to a processing unit (120). The processing unit (120) may include a processor (404) operatively coupled to a memory (406). The memory (406) may store one or more calibration parameters and executable instructions. When executed by the processor (404), the instructions may cause the processing unit (120) to receive the plurality of 2-D ultrasound frames (114) from the linear ultrasound probe (102). The processor (404) may also be configured to identify, in each 2-D ultrasound frame (114), a reflector trace corresponding to a linear acoustic reflector (110) disposed in a slanted, non-parallel orientation within an inner scan track (108) of the reflector-integrated mask (RIM) (104). The processor (404) may also be configured to determine a relative lateral position of each 2-D ultrasound frame (114) based on the reflector trace and the calibration parameter. The processor (404) may also be configured to generate a 3-D volumetric reconstruction (116) of the target anatomy using the plurality of 2-D ultrasound frames (114) and their respective relative lateral positions.

[0060] In one embodiment, the processing unit (120) may generate the 3-D volumetric reconstruction (116) by spatially arranging the plurality of 2-D ultrasound frames (114) according to their respective relative lateral positions determined from the reflector trace (118). Because each 2-D ultrasound frame (114) corresponds to a distinct physical position along the scan direction (Y), the processing unit (120) may map each frame into a common coordinate space, such that each imaging plane (112) occupies a unique location within the volumetric field. The plurality of 2-D ultrasound frames (114) may then be assembled sequentially to form a volumetric representation of the target anatomy. In some implementations, slice-to-slice interpolation may be performed along the scan direction (Y) to regularize sampling density or improve anatomical continuity without modifying the fundamental frame-wise acquisition structure.

[0061] In another embodiment, the processing unit (120) may perform intensity-based compounding or voxel-level resampling to populate a volumetric grid using pixel values from adjacent 2-D ultrasound frames (114). The pixel depths measured along the axial direction (Z) may be projected into corresponding spatial locations while maintaining the lateral (X) and translational (Y) coordinates calculated for each frame. Standard ultrasound imaging algorithms, such as delay-and-sum-based spatial mapping, may be applied on a per-frame basis to convert raw channel data into B-mode intensity distributions that are subsequently fused into a unified volumetric domain. The output may be visualized as 3-D intensity maps, isosurfaces, or tomographic slices oriented along arbitrary planes, providing enriched anatomical context compared to individual 2-D ultrasound frames (114).

[0062] In one embodiment, the memory (406) of the processing unit (120) may store a calibration parameter that includes a reflector angle and a reflector offset determined during an initial two-point static calibration of the reflector-integrated mask (RIM) (104). For the two-point static calibration, the linear ultrasound probe (102) may be positioned successively at a first reference position (A) and a second reference position (B) (shown in FIG. 2) along opposite ends of the inner scan track (108) while the probe remains stationary at each position. At each position, the imaging plane (112) may acquire a 2-D ultrasound frame (114) and the reflector trace (118) corresponding to the linear acoustic reflector (110) may be detected. The pixel-depths of the reflector trace (118) measured at the two reference positions may be recorded, and a closed-form geometric relationship may be solved to obtain the reflector angle relative to the imaging plane (112) and a reflector offset representing a depth intercept associated with the linear acoustic reflector (110).

[0063] In another embodiment, once the reflector angle and the reflector offset are computed from the two measured pixel-depths, these values may be stored in the memory (406) as the calibration parameter and subsequently used by the processor (404) to determine a relative lateral position for each subsequently acquired 2-D ultrasound frame (114) during scanning. Because the two-point static calibration is performed without probe motion and uses the fixed geometry of the inner scan track (108), the resulting reflector angle and reflector offset may remain stable across sessions, thereby enabling consistent self-localization and accurate generation of the 3-D volumetric reconstruction (116).

[0064] Referring now to FIG. 5, an exemplary schematic diagram of the RIM (104) in Y-axis direction, in accordance with one or more embodiment of the present disclosure. In one embodiment, the reflector-integrated mask (RIM) (104) may include a rigid, four-walled, open-ended rectangular housing having length, height, width, and wall thickness selected to receive and laterally constrain the linear ultrasound probe (102). A vertical clearance ΔH of the inner scan track (108) may be selected to accommodate the ultrasound sensor array (202) and allow unconstrained movement of the linear ultrasound probe (102) from one end of the inner scan track (108) to the other.

[0065] The reflector-integrated mask (RIM) (104) may further include a linear acoustic reflector (110) integrated into or fixed to a side wall of the inner scan track (108). As shown in FIG. 5(a), the linear acoustic reflector (110) may extend along the scan direction (Y) with a defined reflector length (L_R) and a protrusion width W_R (shown in FIG. 3). In one embodiment, the protrusion width W_R may be chosen as an integer multiple of the element size of the ultrasound sensor array (202), which may promote consistent reflector visibility in the imaging plane (112) across B-mode acquisitions. The linear acoustic reflector (110) may be disposed in a slanted, non-parallel orientation with respect to the imaging plane (112), beginning at a first depth d_A at one end (S_A) of the inner scan track (108) and ending at a second depth d_B at the opposite end (S_B). The linear acoustic reflector (110) may intersect intermediate positions S_N along the scan direction (Y), giving rise to a reflector trace (118) at an intermediate pixel-depth d_N in the corresponding 2-D ultrasound frame (114).

[0066] With reference to FIG. 5(a), the inner scan track (108) may define a substantially linear translation path between positions S_B and S_A. Because the linear acoustic reflector (110) is inclined by an angle θ relative to the imaging plane (112), B-mode ultrasound imaging at different positions along the scan direction (Y) may yield different reflector depths. The difference in depth values (e.g., d_A, d_N, d_B) across successive ultrasound frames may encode the relative lateral displacement of the linear ultrasound probe (102). In this way, the reflector trace (118) observed in each 2-D ultrasound frame (114) may serve as an internal positional reference, without requiring external mechanical encoders or tracking sensors.

[0067] FIGS. 5(b)–5(d) illustrate example 2-D ultrasound frames (114) collected at different positions along the scan direction (Y). At position S_B, the reflector trace (118) may appear at a depth d_B. At an intermediate position S_N, the same reflector trace (118) may appear at an intermediate depth d_N. At the opposite end position S_A, the reflector trace (118) may appear at a depth d_A. In certain embodiments, known image-to-distance conversion factors (e.g., axial pixel size) in combination with the reflector inclination angle θ may be used to determine the relative lateral position of each 2-D ultrasound frame (114). The difference in reflector depth across positions may therefore provide a reliable closed-form means for mapping each frame to its spatial coordinate along the scan direction (Y).

[0068] The distance along the scan direction (Y) corresponding to a measured reflector depth d_N may be represented in FIG. 5(a) as L_N, and the axial offset between positions S_A and S_B may be denoted by D. The height H of the reflector-integrated mask (RIM) (104) may define the enclosure profile around the scanning axis. In this configuration, the apparent reflector depth in each image slice is determined by both the reflector offset and the reflector angle θ, enabling the processing unit (120) to back-calculate probe location from the geometry.

[0069] In one embodiment, three-dimensional (3-D) imaging using the reflector-integrated mask (RIM) (104) may begin by positioning the reflector-integrated mask (RIM) (104) on subject skin (206) overlying a target anatomy (204). The linear ultrasound probe (102) may be placed adjacent to a first end (S_A) of the inner scan track (108) and then translated toward an opposite end (S_B) along the scan direction (Y), while acquiring a plurality of two-dimensional (2-D) ultrasound frames (114). Raw image data associated with each position along the inner scan track (108) may be stored by the processing unit (120) for subsequent generation of the 3-D volumetric reconstruction (116). Each 2-D ultrasound frame (114) obtained at successive probe locations between S_A and S_B may later be spatially stitched based on an internally tracked depth encoding of a linear acoustic reflector (110).

[0070] The relative position of each 2-D ultrasound frame (114) along the scan direction (Y) may be determined based on the reflector trace (118) of the linear acoustic reflector (110) recorded within that frame. Because the linear acoustic reflector (110) is oriented at an inclination angle θ with respect to the imaging plane (112), its depth location in the axial direction (Z) changes as the linear ultrasound probe (102) translates along the scan direction (Y). Thus, each 2-D ultrasound frame (114) corresponds to a unique reflector depth. As illustrated in FIG. 5(c), the depth of the reflector trace (118) in an intermediate frame may be expressed as:

$$d_N = D + d \qquad (1)$$

where d_N is the measured reflector depth in the n-th frame, D is a fixed geometric offset associated with the reflector-integrated mask (RIM) (104), and d is a variable component determined by the reflector's slanted position. The depth d_N may further be computed from the pixel index of the reflector trace (118). Let P_n denote the n-th pixel index at which the reflector trace (118) is observed along the depth axis, and let d_Z represent the pixel-to-depth conversion factor. The depth d_N may then be computed as:

$$d_N = P_n + \ d_Z \qquad (2).$$

[0071] Once the reflector depth d_N is identified, the relative lateral position L_N of the corresponding 2-D ultrasound frame (114) along the scan direction (Y) may be calculated using the geometry of the reflector-integrated mask (RIM) (104). In one embodiment, the relative lateral position L_N may be determined as:

$$L_N = \frac{d}{\tan\theta} = \frac{d_N - D}{tan\,\theta} \qquad (3)$$

[0072] Here, D and $tan\theta$ may be constants specific to a particular reflector-integrated mask (RIM) (104), while d_N may be derived from the corresponding 2-D ultrasound frames (114). The processing unit (120) may compute the relative lateral positions. L_N for each of the 2-D ultrasound frames (114), and may subsequently use these positions together with the stored 2-D ultrasound frames (114) to generate the 3-D volumetric reconstruction (116).

[0073] In one embodiment, when the linear ultrasound probe (102) is translated from a first end to a second end of the inner scan track (108), a set of N two-dimensional (2-D) ultrasound frames (114) may be generated. For each 2-D ultrasound frame (114), the processing unit (120) may apply a conventional delay-and-sum (DAS) beamforming algorithm to obtain a B-mode image, and may optionally perform a Hilbert-transform-based envelope detection to emphasize intensity profiles. Within each B-mode image, the reflector trace (118) produced by the linear acoustic reflector (110) may be detected as a local intensity peak. The reflector depth d_N for the n-th frame may be computed from the detected pixel location, and the corresponding relative lateral position L_N may then be determined as previously described using the stored calibration parameter. After positions are computed, frames captured at substantially identical positions may be identified and averaged to reduce noise, and the remaining frames may be reordered in ascending L_N so that the data set is spatially consistent despite transducer-to-transducer variations.

[0074] The 3-D volumetric reconstruction (116) may be generated by stacking the reordered 2-D ultrasound frames (114) according to their relative lateral positions L_N and resampling the intensity values onto a volumetric grid. To isolate anatomy of interest, a region-of-interest (RoI) may be delineated interactively (for example, using a polygonal selection), followed by creation of a binary mask for the selected region. Morphological operations such as closure may be applied to fill small gaps and smooth the mask. The processing unit (120) may compute statistical measures (e.g., mean and standard deviation of RoI intensities) to derive an adaptive threshold that accommodates depth-dependent intensity variations. The segmented volume may then be rendered and reviewed using a volume visualization tool, enabling interactive slicing and detailed examination of the reconstructed data while providing precise spatial registration between the volume and the underlying plurality of 2-D ultrasound frames (114).

[0075] In one embodiment, the 3-D volumetric reconstruction (116) described previously may assume ideal geometric placement of the linear acoustic reflector (110) and accurate calibration of the reflector-integrated mask (RIM) (104). In practical scenarios, however, various non-idealities may introduce error in positional mapping of the plurality of two-dimensional (2-D) ultrasound frames (114). These non-idealities may include (i) deviation in the fixed offset D and the reflector angle , (ii) generation of multiple ultrasound frames at substantially identical physical locations along the scan direction (Y), and (iii) probe scans initiated at arbitrary positions along the inner scan track (108).

[0076] Manufacturing tolerances associated with the reflector-integrated mask (RIM) (104) may introduce deviations in the fixed offset D or the designed reflector angle θ. A deviation in D, expressed as ΔD , may result in a constant offset error ΔL_N applied to all calculated frame positions L_N, given by:

$$\Delta L_N = \pm(\frac{\Delta D}{tan\theta}) \quad (4).$$

[0077] Because this deviation is uniformly added to all calculated positions L_N, it may not cause distortion in the spatial stacking of the 2-D ultrasound frames (114). Likewise, a deviation in θ may cause scaling of all L_N values without altering their ordering. Although this does not affect correct frame positioning, the resulting scale change may impact volumetric measurements derived from the 3-D volumetric reconstruction (116).

[0078] To mitigate the scaling effect caused by deviation in θ, a one-time static calibration may be performed. The linear ultrasound probe (102) may be held stationary at the first end (S_A) of the inner scan track (108) to obtain a first reflector depth d_A, and subsequently at the opposite end (S_B) to obtain a second reflector depth d_B. Because the physical distance L_{AB} between S_A and S_B is known, the actual reflector angle θ may then be determined as:

$$\theta = tan^{-1}(\frac{d_A - d_B}{L_{AB}}) \quad (5)$$

[0079] Once computed, the updated reflector angle θ may be stored in memory (406) and used by the processing unit (120) to correct positional estimations for subsequent scans.

[0080] In some embodiments, the system (100) may generate multiple 2-D ultrasound frames (114) for the same location along the scan direction (Y), particularly when the mechanical scan speed is slower than the ultrasound frame-capture rate. These redundant frames may be averaged by the processing unit (120) to form a single composite frame for each location, thereby improving signal quality and mitigating noise. Because position estimation for each 2-D ultrasound frame (114) is based on reflector depth, initiating a scan at any location between S_A and S_B does not affect accuracy of subsequent 3-D volumetric reconstruction (116).

[0081] Motion-induced errors may occur when the linear ultrasound probe (102) is moved at a velocity exceeding the spatial sampling rate determined by the frame rate F_r. To avoid frame skipping, the scanning velocity v may satisfy:

$$v \leq (\Delta L_{min} \times F_r) \quad (6)$$

[0082] where ΔL_{min} represents the minimum resolvable lateral displacement dictated by system sensitivity. Since the reflector depth d_N corresponds to a pixel index multiplied by depth resolution d_z, the minimum resolvable displacement ΔL_{min} may be expressed as:

$$\Delta L_{min} = \frac{d_z}{\tan\theta} \quad (7)$$

[0083] In one implementation, using system parameters described for the prototype, the maximum allowable scan velocity without introducing motion-related artifacts may be approximately 2112.2 mm/s. During use, clinicians may ensure that the probe translation remains below this threshold to preserve reconstruction fidelity and minimize distortion in the 3-D volumetric reconstruction (116).

[0084] In one embodiment, the processing unit (120) may be configured to compute a relative lateral position of each of the plurality of two-dimensional (2-D) ultrasound frames (114) along the scan direction (Y) using a closed-form geometric relationship derived from the pixel-depth of a reflector trace captured in each 2-D ultrasound frame (114). For each 2-D ultrasound frame (114), the reflector trace corresponding to the linear acoustic reflector (110) may be identified, and its pixel-depth within the imaging plane (112) may be extracted. Using this pixel-depth, the processing unit (120) may determine a lateral position value that represents the location at which the given 2-D ultrasound frame (114) was acquired within the reflector-integrated mask (RIM) (104).

[0085] In one embodiment, the processing unit (120) may be configured to order the plurality of two-dimensional (2-D) ultrasound frames (114) according to their respective relative lateral positions along the scan direction (Y) prior to generation of the three-dimensional (3-D) volumetric reconstruction (116). After computing a relative lateral position L_N for each 2-D ultrasound frame (114) using the pixel-depth of the reflector trace (118) and the stored calibration parameter, the processing unit (120) may sort the plurality of 2-D ultrasound frames (114) in ascending (or descending) order of L_N. The resulting ordered sequence may establish a consistent spatial progression of imaging planes (112) across the scan direction (Y).

[0086] In one embodiment, when two or more 2-D ultrasound frames (114) share substantially identical relative lateral positions, the processing unit (120) may optionally average intensity values or select a representative frame to minimize redundancy while preserving spatial fidelity. The ordered (and optionally consolidated) sequence may then be supplied to subsequent reconstruction operations so that the 3-D volumetric reconstruction (116) is assembled in the correct spatial order derived from the reflector-based localization.

[0087] In one embodiment, the processing unit (120) may generate the 3-D volumetric reconstruction (116) using delay-and-sum (DAS) beamforming applied to the plurality of two-dimensional (2-D) ultrasound frames (114). For each acquisition position along the scan direction (Y), channel data from the linear ultrasound probe (102) may be time-aligned using propagation delays calculated for sample points within the imaging plane (112), and coherently summed to form a B-mode intensity image. The processing unit (120) may optionally perform envelope detection (e.g., Hilbert-transform-based) and dynamic range mapping to obtain a display-ready B-mode frame for that position.

[0088] After DAS beamforming produces the set of B-mode 2-D ultrasound frames (114), the processing unit (120) may associate each frame with its relative lateral position computed from the reflector trace (118) and the stored calibration parameter, order the frames by position, and assemble the ordered frames into the volumetric domain to yield the 3-D volumetric reconstruction (116). In some implementations, voxel resampling and slice-to-slice interpolation may be applied to regularize spacing along the scan direction (Y) while preserving intensity relationships produced by the DAS beamforming stage.

[0089] In one exemplary embodiment, the reflector-integrated mask (RIM) may be constructed and operated using the exemplary parameter values summarized in Table I given below.

Parameters	Symbol	Value
RIM length	L	120 mm
RIM width	W	60 mm
RIM height	H	30 mm
RIM thickness	T	20 mm

RIM offset depth	D	5 mm
Scan track height	ΔH	5 mm
Metal reflector length	L_R	100 mm
Metal reflector width	W_R	20 mm
Metal reflector thickness	T_R	1 mm
Metal reflector protrusion	ΔW	2 mm
Metal reflector inclination angle	θ	5°
Tx center frequency	f	7.60 MHz
No. of elements in ultrasound (US) probe	Nc	128
Length of US probe	L_{US}	38.40 mm
Type of transmission	Tx	Plane wave
Frame rate	Fr	10,000 fps

Table I

[0090] In one exemplary embodiment, the reflector-integrated mask (RIM) (104) may be constructed to include a linear track configured to accommodate translation of various linear ultrasound probes along a scan direction. The geometric parameters of the RIM, such as its length, width, height, thickness, and acoustic reflector placement, may be selected based on factors including probe dimensions, target anatomy, clinical workflow, and ergonomic preference. Similarly, the linear metallic reflector may be dimensioned according to the characteristics of the imaging plane and the desired reflector visibility within acquired ultrasound frames. The present disclosure is not limited to any particular physical dimensions, and the selection of suitable dimensions lies within the scope of ordinary design choice.

[0091] If included, representative parameter values are provided only as non-limiting examples to illustrate one possible configuration of the reflector-integrated mask and compatible ultrasound probe. Other geometries, dimensions, and probe specifications may be used without departing from the scope of the present disclosure.

[0092] Referring to FIG. 6, an exemplary block diagram of the processing unit (120) is illustrated, in accordance with one or more embodiments of the present disclosure. The processing unit (120) may be configured to receive a plurality of two-dimensional (2-D) ultrasound frames (114) acquired by the linear ultrasound probe (102), identify reflector traces corresponding to a linear acoustic reflector (110), determine relative lateral positions of the plurality of 2-D ultrasound frames

(114), and generate a three-dimensional (3-D) volumetric reconstruction (116) of a target anatomy. In various deployments, the processing unit (120) may be incorporated within a workstation, integrated in a bedside imaging console, or operatively coupled to the linear ultrasound probe (102) through a wired or wireless communication link.

[0093] In one embodiment, the processing unit (120) may include one or more processors (602). The one or more processors (602) may be implemented using microprocessors, microcontrollers, digital signal processors, graphics processing units, application-specific integrated circuits, system-on-chip devices, or combinations thereof capable of executing computer-readable instructions.

[0094] The processing unit (120) may further include a main memory (604), a read-only memory (606), and a mass storage device (608). The main memory (604) may provide volatile storage (e.g., random access memory (RAM)) for maintaining temporary frame buffers, intermediate positional estimates, and active reconstruction pipelines. The read-only memory (606) may store secure boot firmware, initialization code, and calibration parameters including reflector angle and reflector offset. The mass storage device (608) may provide non-volatile storage (e.g., flash or solid-state memory) for the plurality of 2-D ultrasound frames (114), historical imaging session records, exported 3-D volumetric reconstruction (116), and application-level configuration data. These components may communicate with the one or more processors (602) via an internal interconnect bus (612).

[0095] The processing unit (120) may expose one or more communication port(s) (610) to facilitate bidirectional exchange of data with external components. The communication port(s) (610) may support interfaces such as Universal Serial Bus, Ethernet, or serial communication to provide stable connectivity for streaming the plurality of 2-D ultrasound frames (114) and for transmitting reconstructed 3-D volumetric data for visualization or archival.

[0096] An external storage device (614) may be operatively coupled to the processing unit (120) to enable import of calibration configuration profiles, reflector-parameter maps, or updated software images, and to export reconstructed

3-D data sets for remote review. The external storage device (614) may include removable solid-state media or network-attached storage, depending on clinical workflow requirements.

[0097] A user interface (616) may be provided for local operator interaction. The user interface (616) may allow an operator to initiate frame acquisition, select reconstruction modes, inspect reflector-tracking outputs, or review volumetric reconstruction results. The user interface (616) may include a graphical display, a touchscreen terminal, or mechanical input controls, depending on system configuration.

[0098] The block configuration illustrated in FIG. 6 is exemplary. Components within the processing unit (120) may be consolidated or distributed across multiple computing modules, and additional components may be included or omitted without departing from the scope of the present disclosure.

[0099] FIG. 7 illustrates an exemplary flow diagram of a method (700) for three-dimensional (3-D) ultrasound volumetric reconstruction, in accordance with one or more embodiments of the present disclosure. The method (700) may be performed using the system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction described herein.

[00100] At step (702), the method (700) may include acquiring, by the linear ultrasound probe (102), a plurality of two-dimensional (2-D) ultrasound frames (114) of a target anatomy. The acquiring may include constrained translation of the linear ultrasound probe (102) within an inner scan track (108) of a reflector-integrated mask (RIM) (104), along a scan direction (Y) of the target anatomy.

[00101] At step (704), the method (700) may include receiving, by a processor (404) operatively coupled to a memory (406), the plurality of 2-D ultrasound frames (114) acquired along the scan direction (Y). The receiving may include transfer of captured data to a processing unit (120) for subsequent computational analysis.

[00102] At step (706), the method (700) may include identifying, by the processor (404), in each 2-D ultrasound frame (114), a reflector trace (118) corresponding to a linear acoustic reflector (110) of the reflector-integrated mask (RIM) (104). The linear acoustic reflector (110) may be disposed in a slanted, non-parallel orientation with respect to an imaging plane (112) of the linear ultrasound probe (102), such that the reflector trace (118) remains observable within the field-of-view during probe translation.

[00103] At step (708), the method (700) may include accessing, by the processor (404), a calibration parameter stored in the memory (406). The calibration parameter may include reflector orientation and offset information used to determine a spatial position of each 2-D ultrasound frame (114) along the scan direction (Y).

[00104] At step (710), the method (700) may include determining, by the processor (404), based on the reflector trace (118) and the calibration parameter, a relative lateral position of each 2-D ultrasound frame (114) along the scan direction (Y). The relative lateral position may correspond to a mapped frame location used to organize the plurality of 2-D ultrasound frames (114) spatially.

[00105] At step (712), the method (700) may include generating, by the processor (404), using the plurality of 2-D ultrasound frames (114) and their respective relative lateral positions, a three-dimensional (3-D) volumetric reconstruction (116) of the target anatomy. The generating may include spatial ordering of the plurality of 2-D ultrasound frames (114) to reconstruct a volumetric representation for visualization or diagnostic interpretation.

[00106] It may be appreciated that one or more steps of the method (700) may be performed in an alternate sequence, combined, or supplemented with other steps, without departing from the scope of the present disclosure.

[00107] Therefore, the system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction provides sensorless probe tracking to determine the spatial position of the linear ultrasound probe (102) without the need for separate external tracking sensors. By relying on the linear acoustic reflector (110) integrated within the reflector-integrated mask (RIM) (104) and the corresponding reflector trace (118) identified in each two-dimensional (2-D) ultrasound frame (114), the system (100) may minimize mechanical complexity and reduce operational failure points normally associated with moving or electromechanical tracking elements. This sensorless approach may support robust image acquisition while simplifying maintenance requirements in clinical workflows.

[00108] The system (100) may further support compatibility with conventional imaging devices, as the system (100) may utilize a standard two-dimensional (2-D) ultrasound probe (102) together with conventional delay-and-sum (DAS) beamforming to reconstruct B-mode images. This compatibility may facilitate seamless integration with existing ultrasound equipment, enabling straightforward adoption without requiring specialized hardware. Additionally, by combining sensorless probe tracking with established imaging techniques, the system (100) may achieve low implementation complexity and operational ease. The reflector-integrated mask (RIM) (104) may be lightweight, enabling high portability and supporting a cost-effective imaging workflow suited for diverse clinical environments, including resource-constrained settings. Collectively, these advantages may position the system (100) as an efficient, practical, and affordable solution for volumetric imaging.

[00109] While the foregoing describes example embodiments, other and further embodiments may be devised without departing from the scope defined by the claims, and the embodiments, versions, and examples are non-limiting and are provided solely to enable a person having ordinary skill in the art to make and use the present disclosure in view of their general knowledge.

ADVANTAGES OF THE PRESENT DISCLOSURE

[00110] The present disclosure provides a system and a method for achieving three-dimensional (3-D) ultrasound volumetric reconstruction of a target anatomy using a conventional two-dimensional (2-D) ultrasound probe without reliance on external tracking devices.

[00111] The present disclosure provides sensorless localization enabled by the reflector-integrated mask, allowing determination of relative frame positions based on a reflector trace within acquired ultrasound frames.

[00112] The present disclosure provides reduced system complexity by eliminating the need for optical, electromagnetic, or mechanical tracking hardware, thereby improving portability, reliability, and ease of maintenance.

[00113] The present disclosure provides compatibility with standard ultrasound imaging pipelines, including delay-and-sum (DAS) beamforming, facilitating seamless integration with existing clinical workflows.

[00114] The present disclosure provides cost-effective 3-D imaging capability suitable for diverse clinical environments, including resource-constrained settings.

[00115] The present disclosure provides a compact reflector-integrated mask architecture that supports predictable probe translation for generating spatially organized volumetric datasets.

We Claim:

1. A reflector-integrated mask (RIM) (104) for ultrasound imaging, comprising:

an outer scan track (106) defined by a rigid, four-walled, open-ended rectangular cuboid housing;

an inner scan track (108) disposed within the outer scan track (106), wherein the inner scan track (108) is dimensioned to receive a linear ultrasound probe (102) and to constrain translation of the linear ultrasound probe (102) along a scan direction; and

a linear acoustic reflector (110) fixed to or embedded in a side wall of the inner scan track (108) and extending along the scan direction, wherein the linear acoustic reflector (110) is disposed in a slanted, non-parallel orientation with respect to an imaging plane (112) of the received linear ultrasound probe (102) and positioned to lie within a field of view of the imaging plane (112) during the translation.

2. The RIM (104) as claimed in claim 1, wherein the linear acoustic reflector (110) comprises at least one of metallic object, plastic object or polylactic acid (PLA) based object or a combination thereof.

3. The RIM (104) as claimed in claim 1, further comprising an acoustic-coupling medium disposed within the inner scan track (108).

4. The RIM (104) as claimed in claim 1, wherein a protrusion width (W_R) of the linear acoustic reflector (110) is an integer multiple of an element size of the linear ultrasound probe (102).

5. A system (100) for three-dimensional (3-D) ultrasound volumetric reconstruction, comprising:

a linear ultrasound probe (102) configured to acquire a plurality of two-dimensional (2-D) ultrasound frames (114) of a target anatomy;

a reflector-integrated mask (RIM) (104) comprising:

an inner scan track (108) dimensioned to receive the linear ultrasound probe (102) and to constrain translation of the linear ultrasound probe (102) along a scan direction of the target anatomy; and

a linear acoustic reflector (110) fixed to or embedded in a side wall of the inner scan track (108) and extending along the scan direction, wherein the linear acoustic reflector (110) is disposed in a slanted, non-parallel orientation with respect to an imaging plane (112) of the linear ultrasound probe (102) and positioned to lie within a field of view of the imaging plane (112) during said translation;

a memory (406);

a processor (402) operatively coupled to the linear ultrasound probe (102) and the memory (406), wherein the memory (406) stores instructions that when executed by the processor (402), cause the processor (402) to:

receive, from the linear ultrasound probe (102), the plurality of 2-D ultrasound frames (114) acquired along the scan direction;

identify, in each 2-D ultrasound frame (114), a reflector trace (118) corresponding to the linear acoustic reflector (110);

determine, based on the reflector trace (118) and a calibration parameter stored in the memory (406), a relative lateral position of each 2-D ultrasound frame (114) along the scan direction; and

generate, using the plurality of 2-D ultrasound frames (114) and the respective lateral positions, the 3-D volumetric reconstruction (116) of the target anatomy.

6. The system (100) as claimed in claim 5, wherein the stored calibration parameter comprises a reflector angle and a reflector offset determined during initial two-point static calibration.

7. The system (100) as claimed in claim 5, wherein the processor (402) is configured to compute the relative lateral position of each 2-D ultrasound frame

(114) according to a closed-form geometric relationship based on the pixel-depth of the reflector trace (118).

8. The system (100) as claimed in claim 5, wherein the processor (402) is configured to order the plurality of 2-D ultrasound frames (114) based on their respective relative lateral positions along the scan direction prior to generation of the 3-D volumetric reconstruction (116).

9. The system (100) as claimed in claim 5, wherein the processor (402) is configured to generate the 3-D volumetric reconstruction (116) using delay-and-sum (DAS) beamforming applied to the plurality of 2-D ultrasound frames (114).

10. A method (700) for three-dimensional (3-D) ultrasound volumetric reconstruction, the method (700) comprising:

acquiring (702), by the linear ultrasound probe (102), a plurality of two-dimensional (2-D) ultrasound frames (114) of a target anatomy by constrained translation of the linear ultrasound probe (102), within an inner scan track (108) of a reflector-integrated mask (RIM) (104), along a scan direction of the target anatomy;

receiving (704), by a processor (402) operatively coupled to a memory (406), the plurality of 2-D ultrasound frames (114) acquired along the scan direction;

identifying (706), by the processor (402), in each 2-D ultrasound frame (114), a reflector trace (118) corresponding to the linear acoustic reflector (110) of the RIM (104) positioned to lie within a field of view of an imaging plane (112) of the linear ultrasound probe (102) during the translation, wherein the linear acoustic reflector (110) is disposed in a slanted, non-parallel orientation with respect to the imaging plane (112);

accessing (708), by the processor (402), a calibration parameter stored in the memory (406);

determining (710), by the processor (402), based on the reflector trace (118) and the calibration parameter, a relative lateral position of each 2-D ultrasound frame (114) along the scan direction; and

generating (712), by the processor (402), using the plurality of 2-D ultrasound frames (114) and their respective relative lateral positions, a 3-D volumetric reconstruction (116) of the target anatomy.

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ABSTRACT

SYSTEM AND METHOD FOR 3D ULTRASOUND IMAGING OF TARGET ANATOMY

A system (100) and a method (700) for three-dimensional (3-D) ultrasound volumetric reconstruction is disclosed. The system (100) includes a linear ultrasound probe (102) configured to acquire a plurality of two-dimensional (2-D) ultrasound frames (114) of a target anatomy. A reflector-integrated mask (RIM) (104) includes an inner scan track (108) dimensioned to receive the linear ultrasound probe (102) and constrain translation along a scan direction, and a linear acoustic reflector (110) fixed to or embedded in a side wall of the inner scan track (108). The linear acoustic reflector (110) is slanted relative to an imaging plane (112) of the linear ultrasound probe (102) to ensure visibility within the field-of-view during translation. A processor (402) accesses a calibration parameter stored in a memory (406), identifies a reflector trace (118) in each 2-D ultrasound frame (114), determines relative lateral frame positions, and generates a 3-D volumetric reconstruction (116) of the target anatomy.

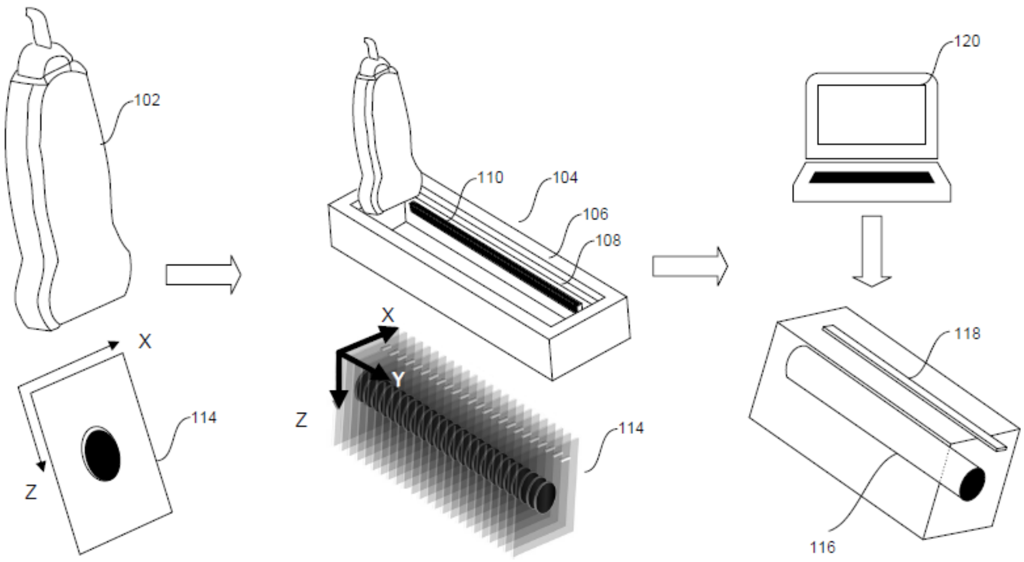

FIG. 1